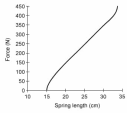


The relationship between the length of a spring and the force required to stretch the spring is shown below.



What is the work done when the spring length is increased from 20 cm to 30 cm?

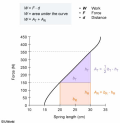
- ☒ A. 25 J [80%]
- ☐ B. 35 J [15%]
- ☐ C. 250 J [20%]
- ☐ D. 350 J [25%]

Correct

80% answered correctly

Explanation:

Work from area under the curve (force vs distance)



Answer:

Mechanical work refers to the transfer of energy in or out of a system that occurs as the result of the application of a force (F) over some distance (d):

$$W = F \cdot d$$

The **SI unit** for work is the joule (J), where 1 J is equal to 1 newton (N) of force exerted over 1 meter (m) of distance (1 J = 1 N·m).

In a graph of force versus distance, the **area under the curve** is equal to work associated with the displacement because the $F \cdot d$ component of the above equation is equal to a y-axis value (a force) multiplied by an x-axis value (a distance).

In this question, the graph shows a **variable force** being applied to the end of a spring to stretch it from a length of 15 cm to a length of 35 cm. When stretching the spring from 20 cm to 30 cm long, the variable force acts over a distance of 10 cm. **Converting this distance to standard units** results in:

$$d = \left(10 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} \right) = 0.10 \text{ m}$$

In the graph of force versus distance, the area under the curve can be calculated by visualizing the total area as the sum of the **areas (A)** of two geometric shapes (one triangle and one rectangle) that both have a base 0.10 m in length. The combined area of these two shapes yields the work:

$$\begin{aligned} \text{work} &= (\text{rectangle area}) + (\text{triangle area}) \\ W &= b_R \cdot h_R + \frac{1}{2} \cdot b_T \cdot h_T \\ W &= (0.10 \text{ m}) \cdot 150 \text{ N} + \frac{1}{2} (0.10 \text{ m}) \cdot 300 \text{ N} \\ W &= 15 \text{ J} + 15 \text{ J} = 30 \text{ J} \end{aligned}$$

Therefore, stretching the spring from a length of 20 cm to 30 cm through the application of a variable force is associated with mechanical work equal to 25 J in magnitude (**Choice A**).

Educational objective:

Work describes the energy transfer into or out of a system that takes place as the result of a force acting through a distance. In a graph of force versus distance, the work associated with a given displacement can be derived from the graph by evaluating the total area under the curve.

Time applied to:
Subject:
Physics

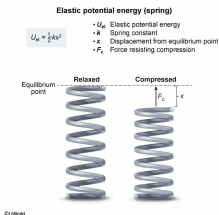
QED-102421
System:
Mechanics and Energy

A 1-m spring is compressed to half its original length. What is the potential energy stored within the spring? (Note: $k = 40 \text{ N/m}$)

- ☐ A. 2.5 J [1%]
- ☒ B. 5.0 J [56%]
- ☐ C. 10 J [14%]
- ☐ D. 20 J [27%]

Correct
55% answered correctly

Explanation:



Potential energy refers to a form of stored energy that depends on the spatial position or configuration of objects. Essentially, potential energy is the capacity of an object within a system to do work in the future based on some work completed on it in the past. In mechanical systems, potential energy accumulates because of the **work** performed by a force applied over some distance.

The type of potential energy stored within an object is related to the type of force that previously performed work on the object. For example, **gravitational potential energy** accumulates when the height of an object increases because a force must act against gravity to raise the object.

Similarly, stretching or compressing a spring away from its **equilibrium point** (ie, its position at rest) increases its **elastic potential energy (U_{el})** because displacing a spring requires doing work against the elastic forces resisting displacement. The potential energy stored within a compressed or stretched spring explains why **harmonic oscillations** take place when the spring is released after being compressed or stretched.

The magnitude of potential energy stored within a compressed or stretched spring is a function of spring displacement (x) and the spring constant (k), a measure of the relative elasticity (stiffness) of the spring:

$$U_{el} = \frac{1}{2} k x^2$$

In this question, a spring with a length of 1 m is compressed to 0.5 m by a force, which gives the spring a displacement of 0.5 m. The spring constant is 40 N/m, which indicates that maintaining a displacement of 1 m away from the equilibrium position of the spring requires 40 N of force. Consequently, the magnitude of potential energy stored within the spring from a 0.5-m displacement is:

$$\begin{aligned} U_{el} &= \frac{1}{2} \left(40 \frac{\text{N}}{\text{m}} \right) \cdot \left(0.5 \text{ m} \right)^2 \\ U_{el} &= \frac{1}{2} \left(40 \frac{\text{N}}{\text{m}} \right) \cdot \left(0.25 \text{ m}^2 \right) \\ U_{el} &= 5 \text{ N} \cdot \text{m} = 5 \text{ J} \end{aligned}$$

Therefore, compressing the 1-m spring to half its original length imparts a potential energy into the spring equivalent to 5.0 J (**Choice B**).

Educational objective:

Elastic potential energy refers to the potential energy of springs and other elastic objects. Elastic potential energy increases following the application of a stretching or compressing force that displaces a spring from its equilibrium position.

Time spent:0 sec
Subject:Physics

QID:402482
System:Mechanics and Energy